

Pilot Proposal — Fund for Innovation in Development (FID)

Stage requested: Stage 1 / Pilot grant (up to €200,000)

Project title: FedFraudShield Pilot — Testing privacy-preserving, collaborative fraud detection in 48 Ugandan SACCOs and microfinance institutions

Country: Uganda (ODA-eligible, OECD DAC list)

Applicant: FraudShield Uganda (*legal entity name and registration number to be inserted*)

Lead: Raymond R. Wayesu, Founder & Lead Data Scientist — raymondwayesu@gmail.com · +256 784 902 753

Research partner: (*to be confirmed — see Section 10*)

Duration: 18 months

Requested amount: €198,500

Date: September 2026

Status of this document: working draft prepared alongside the Stage 0 concept note (docs/FID_Stage0_Concept_Note.md). It assumes Stage 0 has been completed; every figure that Stage 0 is meant to produce is marked [Stage 0 result] and must be replaced with actual results. Items marked [TBC] must be confirmed before submission. FID rules (stage definitions, ceilings, eligible costs) must be re-checked against the current call guide on fundinnovation.dev.

1. Executive summary

Internal fraud drains an estimated 2–5% of portfolio value from Uganda's 28,500+ SACCOs and 150+ microfinance institutions every year — UGX 100–250 billion — and when a SACCO fails, its members, mostly rural and low-income households, lose their savings outright. No affordable, data-driven fraud detection exists for this sector: individual institutions have too little data to train a model, and Uganda's Data Protection and Privacy Act (DPPA 2019) prevents pooling raw records.

FedFraudShield solves both constraints at once. Each institution runs a rule-based detector (the FraudShield tool) on its own loan book and, optionally, a small local neural model whose *parameter updates only* — never raw data — are combined into a shared model across institutions. In simulation, federated training raised detection quality (AUPRC) from 0.64 to 0.95. Stage 0 [Stage 0 result: summarise real-data validation with N partner SACCOs – precision/recall on confirmed cases, bandwidth, schema-mapping success rate].

This Stage 1 pilot will deploy the intervention in **48 institutions across four regions for 12 months**, randomising the timing of roll-out so that the pilot generates a credible first estimate of impact on fraud losses and portfolio quality while every participant receives the tool by the end of the pilot. It will answer the questions FID needs answered before a Stage 2 impact evaluation: Does it work on real data at scale? Do SACCO staff act on alerts? What does it cost per institution and per shilling of loss averted? And does federation add value over the rule-based tool alone?

2. Problem and target population

2.1 Who is affected and how

Aspect	Detail
Population	~18 million SACCO/MFI members in Uganda; predominantly rural, informal-sector households. [TBC: cite share of women members; UCSCU/UMRA data]
Direct harm	Loss of savings on SACCO collapse; higher interest and tighter lending to cover losses; exclusion from formal credit when a local SACCO fails.
Sector loss	UGX 100–250 billion / year (≈ USD 27–68 million), on a >UGX 5 trillion asset base.
Institutional failure	PROFIRA (2021): of 453 monitored SACCOs, 312 struggling due to fraud and governance failures; 64 collapsed.
Why existing controls fail	Manual audits sample 5–10% of loans; external audits detect ~3% of fraud (ACFE 2024). Bank-grade fraud tools are priced and designed for institutions with millions of transactions.

2.2 Fraud typologies the intervention targets

Typology	Mechanism	Detection signal
Ghost loans	Loans booked to fictitious borrowers; officer pockets disbursement	Shared phone numbers / IDs across "different" borrowers; missing repayment behaviour
Officer self-lending	Officer approves loans to self or proxies	Officer ID = borrower ID; approvals concentrated in one officer
Loan stacking	Same member takes multiple concurrent loans, often across branches	Same borrower, multiple disbursements same day / overlapping terms
Collusion rings	Officer + group of borrowers + shared guarantor	Shared guarantors, geographic clustering, timing patterns
Timing manipulation	Back-dated or after-hours entries to conceal activity	Approvals outside business hours; end-of-month bunching
Amount manipulation	Inflated disbursements, round-number loans	Statistical outliers vs. product and branch norms

3. The innovation

3.1 Components

Component	Function	Status
FraudShield tool (Layer 1, rule-based)	Ingests any CSV/Excel export; auto-maps columns via schema matching; runs deterministic and statistical rules for the typologies above; produces ranked alerts with evidence. Works from day one with no training data.	Prototyped; deployed in this repository (mvp/, mvp-fast/, backend/). [Stage 0 result: validated on real data at N institutions]

Component	Function	Status
FedFraudShield (Layer 2, federated ML)	Each institution trains a lightweight model locally; differentially-private parameter updates (~100 KB per round) are aggregated on a server hosted in Uganda; the improved global model is returned. No raw data leaves any institution.	Architecture and simulation published (DSA 2026). [Stage 0 result: federated proof-of-concept over real connectivity]
Alert and investigation workflow	Weekly ranked alert list (capped to investigation capacity), SMS/WhatsApp notification, case-tracking sheet, feedback loop that labels confirmed/false cases for model improvement.	Designed; to be finalised in pilot month 1–2.

3.2 What makes it new

- First federated-learning approach to SACCO/MFI fraud in Africa; gives institutions with **no fraud history** a working detector from day one.
- Built for the operating environment: heterogeneous software (Ensibuuko, Kanzu, Excel), 3G connectivity, DPPA 2019 compliance (§14 data minimality, §7(2)(b)(iii) fraud-prevention basis, §19 no cross-border transfer).
- A **deterrence channel** as well as a detection channel: staff who know transactions are monitored change behaviour before any case is proven.

3.3 Evidence to date

Source	Finding
Simulation (PaySim, 15 institutions, non-IID)	Federated AUPRC 0.945 vs. 0.643 local-only (+47%); recall 0.99; 5 zero-history institutions gained detection capability; DP at $\epsilon = 5$ retains AUPRC 0.69.
Stage 0 real-data validation	[Stage 0 result: N institutions, M months of records; % of institution-confirmed fraud cases flagged; false-positive rate at chosen threshold; schema-mapping success on X software systems]
Stage 0 federated PoC	[Stage 0 result: rounds completed, bandwidth, model performance vs local-only on real data]
Stage 0 baselines	[Stage 0 result: median fraud-loss rate, PAR30/90, audit detection rate, cost of investigation per case]

4. Pilot objectives and learning questions

Primary objective: determine whether FedFraudShield reduces fraud losses and improves portfolio quality in real SACCOs at a cost that makes sector-wide adoption viable.

#	Learning question	Answered by
Q1	Does the tool detect real fraud that institutions' existing controls miss, and at what precision?	Alert-level validation; comparison with audit findings
Q2	Do institutions act on alerts? What share are investigated, confirmed, and resolved?	Case-tracking data; process evaluation

#	Learning question	Answered by
Q3	What is the effect on fraud losses, PAR30/PAR90 and write-offs after 12 months?	Phased-roll-out comparison (§6)
Q4	Does federation (Layer 2) add measurable value over the rule-based tool alone (Layer 1)?	Two-arm treatment contrast
Q5	What does it cost per institution, per confirmed case, and per UGX of loss averted?	Cost tracking; cost-effectiveness analysis
Q6	Is there a deterrence effect visible in staff behaviour (after-hours entries, concentration) even before cases are proven?	Behavioural indicators from transaction data
Q7	Are women members and women-led institutions differentially protected?	Gender-disaggregated outcomes
Q8	What is needed for UMRA/UCSCU to operate the federation hub as sector infrastructure?	Governance workstream; stakeholder interviews

5. Pilot sites and sample

Parameter	Design
Number of institutions	48 (40 SACCOs, 8 MFIs) [TBC after Stage 0 power calculation]
Regions	Central (Kampala/Mukono/Wakiso), Eastern (Jinja/Mbale), Western (Mbarara), Northern (Gulu) — 12 per region
Size range	300–5,000 members; 12+ months of digital records; mix of Ensibuuko, Kanza Banking, and Excel/QuickBooks users
Inclusion criteria	UMRA-licensed or UCSCU-affiliated; board consent; a named fraud focal person; exportable loan data
Exclusion criteria	Institutions under active regulatory sanction or receivership; institutions that participated in Stage 0 (they continue separately as demonstration sites)
Recruitment	Through UCSCU regional structures and UMRA [TBC: letters of support]; over-recruit 60 to allow for attrition

6. Pilot design (phased roll-out with randomised timing)

A Stage 1 pilot must test the innovation in real conditions; FID also expects a credible learning design that prepares a Stage 2 impact evaluation. We use a **cluster-randomised phased roll-out** at institution level:

Arm	n	Months 1–3	Months 4–12	Months 13–15
A — Early, full	16	Onboarding	FraudShield tool + federation	Continues
B — Early, rule-based only	16	Onboarding	FraudShield tool only (no federation)	Federation added

Arm	n	Months 1–3	Months 4–12	Months 13–15
C — Delayed	16	Baseline only	Baseline data collection only	Onboarding + full intervention

- **Randomisation:** stratified by region and institution size, conducted by the research partner after baseline data collection; pre-registered (AEA RCT Registry or OSF) [TBC] .
- **Comparison 1 (A + B vs. C, months 4–12):** effect of the intervention on fraud losses and portfolio quality.
- **Comparison 2 (A vs. B):** value added by federation.
- **Fairness:** every institution receives the full intervention by month 15; C-arm institutions receive it free for the remainder of the grant.
- **Power:** with 32 treated vs. 16 control clusters, the pilot is powered to detect large effects on institution-level fraud-loss and PAR outcomes [Stage 0 result: MDE from baseline variance] . It is *not* powered for member-level outcomes — that is the purpose of Stage 2.

7. Implementation plan

Phase	Months	Activities
0. Set-up	1–2	Finalise research partnership and pre-analysis plan; recruit and randomise 48 institutions; sign DPPA-compliant data-processing agreements; ethics approval [TBC: UNCST / institutional REC] ; finalise alert workflow and case-tracking tools; procure local-node hardware.
1. Baseline	2–3	Extract 12–24 months of historical data at all 48 sites; run retrospective detection (results withheld from C-arm until month 13); establish baseline fraud-loss, PAR, audit-detection and investigation-cost measures; staff survey.
2. Onboarding (A, B)	3–4	Install local nodes; validate schema mapping per site; train fraud focal persons and boards (½-day workshop per region); go live with weekly alert cycle.
3. Operation	4–12	Weekly alert cycle at 32 sites; monthly federated training rounds at A-sites; monthly investigation-outcome collection; quarterly threshold tuning; help desk (WhatsApp) and regional field visits every 6 weeks.
4. Roll-out to C; federation to B	13–15	Onboard C-arm; enable federation at B-arm; all 48 sites on full intervention.
5. Analysis and Stage 2 design	13–18	Endline data extraction; impact, process and cost-effectiveness analysis; gender analysis; governance report with UMRA/UCSCU; Stage 2 proposal and full-scale evaluation design.

Alert workflow (per institution, weekly): tool runs on latest export → top-N alerts (N set by investigation capacity, typically 5–10) sent to focal person → focal person records outcome (confirmed fraud / error / false positive / pending) in the case tracker → outcomes feed model recalibration and the evaluation dataset.

8. Evaluation design

8.1 Outcomes

Level	Outcome	Source	Timing
Institution (primary)	Fraud losses as % of portfolio; number and value of confirmed fraud cases	Case tracker; audited accounts	Baseline, month 12, month 15
Institution (primary)	PAR30, PAR90, write-offs	Loan book exports	Monthly
Institution (secondary)	Time from fraud event to detection; investigation cost per case; share of alerts investigated	Case tracker	Continuous
Behavioural (mechanism)	After-hours approvals, officer concentration, round-amount share	Transaction data	Monthly
Institution (secondary)	Solvency indicators; continued licensing; board-reported confidence	UMRA data; board survey	Baseline, endline
Member (exploratory)	Savings retained; loan access; interest rate; trust in institution	Member sample survey (n≈20 per institution)	Baseline, endline
Gender	All above disaggregated by member gender and women-led institutions	As above	As above

8.2 Methods

- **Impact:** intention-to-treat comparison of A+B vs. C over months 4–12 with baseline controls; A vs. B for the federation contrast; difference-in-differences using monthly PAR series as a robustness check.
- **Process:** structured interviews with focal persons and boards at months 6 and 12; alert-log analysis (precision by typology, investigation rates, reasons for non-action).
- **Cost-effectiveness:** full costing of the intervention (development amortised, hardware, connectivity, hosting, training, support, institution staff time) against measured loss reduction and detection gains; comparison with the cost of manual audit per loan covered.
- **Detection accuracy:** precision/recall of alerts against confirmed cases, by typology and institution type; threshold sensitivity.

8.3 Ethics, consent and data protection

- Institutional consent from boards; staff informed that monitoring is in place (this is standard practice and part of the deterrence mechanism); member survey consent individually.
- Raw member data never leaves the institution; the research team receives only aggregate and de-identified alert/outcome data under a data-processing agreement; the aggregation server is hosted in Uganda.
- Alerts are triage inputs for the institution's own investigation, not accusations; disciplinary decisions remain entirely with the institution. A protocol will define how alerts are communicated to avoid unfair treatment of staff.
- Ethics review: [TBC: Uganda National Council for Science and Technology and a registered research ethics committee].

9. Cost-effectiveness plan

Item	Working figure	To be replaced by
Fraud loss per institution	2–5% of portfolio; median SACCO in sample ≈ UGX 500 million portfolio → UGX 10–25 million/yr	Stage 0 baselines; pilot baseline
Intervention cost per institution per year (pilot, fully loaded)	≈ €1,900 (≈ UGX 8 million) including onboarding; ≈ €700 recurring	Actual pilot costs
Break-even	Preventing 30–40% of median losses covers the recurring cost several times over	Measured loss reduction
Comparator	Manual internal audit: 5–10% loan coverage; external audit ≈ UGX 3–8 million/yr for a small SACCO with ~3% fraud detection	[TBC: audit-cost survey]
Metrics to report	Cost per institution-year; cost per confirmed case; cost per UGX 1 million of loss averted; cost per member protected	Endline analysis

10. Team, partners and governance

Role	Who	Responsibilities	Status
Principal Investigator / project lead	Raymond R. Wayesu (MSc Statistics, Linköping; Data Analytics Lead, UVRI; author of FedFraudShield paper)	Overall delivery, detection engine, federation hub	Confirmed
Legal applicant	FraudShield Uganda [TBC: entity, registration, bank account, audited accounts if required]	Grant holder	Must confirm — FID does not fund individuals
Research partner (co-PI for evaluation)	[TBC] — candidates: Makerere University Dept. of Computer Science; a J-PAL-affiliated economist; IPA Uganda	Randomisation, pre-registration, evaluation analysis, publication	To be secured in Stage 0
Field team	1 field coordinator (Kampala) + 4 regional field officers (0.5 FTE each)	Recruitment, onboarding, training, case-tracker collection	To recruit
Engineering	1 ML/backend engineer (1.0 FTE), 1 support engineer (0.5 FTE)	Local nodes, schema mapping, hub operations, help desk	To recruit
Sector partners	Uganda Cooperative Savings and Credit Union (UCSCU); Uganda Microfinance Regulatory Authority (UMRA)	Recruitment channels; governance workstream; pathway to sector-run hub	[TBC: letters of support]
Advisory group	Representatives of UMRA, UCSCU, 2 pilot SACCO boards, research partner, a data-protection lawyer	Quarterly review; ethics and fairness oversight	To convene month 1

Governance: monthly project management meeting; quarterly advisory group; quarterly narrative and financial report to FID; independent evaluation analysis owned by the research partner with publication rights.

11. Budget (18 months) — indicative, EUR

Line	Amount	Basis
Personnel		
Project lead (0.6 FTE × 18 mo)	27,000	Local rate
ML/backend engineer (1.0 FTE × 18 mo)	27,000	
Support engineer (0.5 FTE × 15 mo)	9,000	
Field coordinator (1.0 FTE × 18 mo)	16,200	
Regional field officers (4 × 0.5 FTE × 15 mo)	24,000	
<i>Personnel subtotal</i>	<i>103,200</i>	
Research and evaluation		
Research partner sub-grant (design, randomisation, analysis, publication)	28,000	
Baseline and endline member surveys (48 × 20 × 2 rounds, enumerators, tablets)	9,600	
Ethics review fees, pre-registration, data-protection counsel	3,500	
<i>Research subtotal</i>	<i>41,100</i>	
Deployment		
Local-node hardware (48 × mini-PC/laptop)	14,400	€300 each
Connectivity stipends (48 × 15 mo × €4)	2,900	
Aggregation server hosting in Uganda, security audit, backups	6,000	
SMS/WhatsApp alert gateway	1,500	
<i>Deployment subtotal</i>	<i>24,800</i>	
Training and field operations		
Regional onboarding workshops (4 × 2 rounds)	6,400	Venue, transport, materials
Field travel (6-weekly site visits, 4 regions)	9,000	
Advisory group meetings and final dissemination workshop	3,000	
<i>Training/field subtotal</i>	<i>18,400</i>	
Indirect / contingency		
Audit, accounting, insurance	4,000	

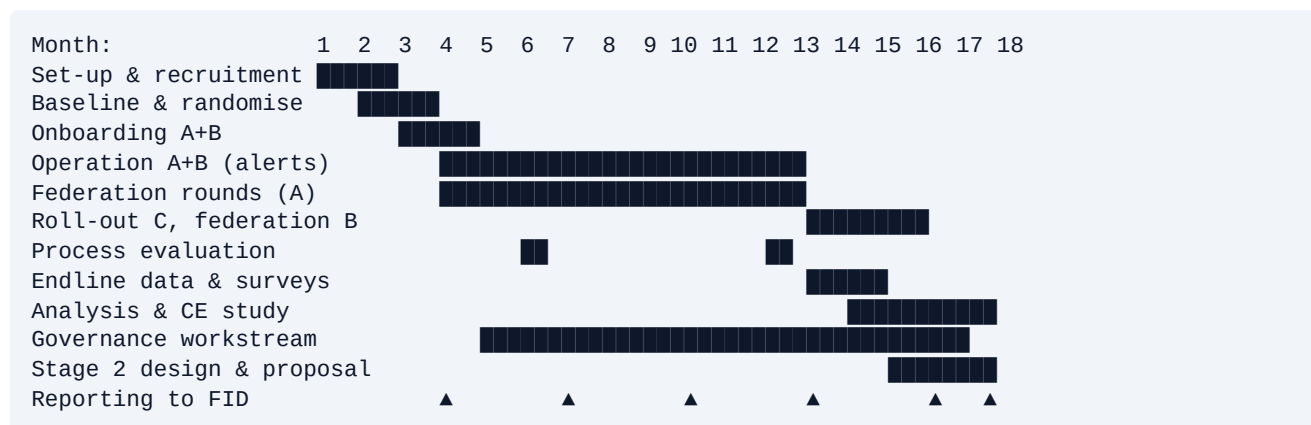
Line	Amount	Basis
Contingency (≈3.5%)	7,000	
Total requested	198,500	

Co-financing: founder's in-kind time and the existing FraudShield codebase (not costed); institutions contribute staff time for investigation and data exports. [TBC: any other co-funding]

12. Risks and mitigations

Risk	Likelihood	Impact	Mitigation
Recruitment shortfall / attrition	Medium	High	Over-recruit to 60; UCSCU/UMRA channels; no-cost participation; C-arm guaranteed the tool by month 13
Institutions do not act on alerts	Medium	High	Cap alerts to capacity; focal-person training; board-level reporting; measure and report non-action as a finding (Q2)
High false-positive rate erodes trust	Medium	Medium	Stage 0 threshold calibration; quarterly re-tuning; typology-specific thresholds; precision tracked per site
Data quality / export failures at Excel-based sites	High	Medium	Schema-mapping module; field-officer support; minimum-data checklist at recruitment
Connectivity failures for federation rounds	Medium	Low	Asynchronous rounds; ~100 KB payloads; rounds can lag a month without harm
Staff intimidation or unfair action on the basis of alerts	Low	High	Alert communication protocol; alerts framed as triage, not accusation; advisory-group oversight; institution retains all decisions
Contamination (C-arm institutions change behaviour after hearing of the pilot)	Medium	Medium	Regional stratification; measure awareness at endline; interpret as lower bound
Regulatory objection to shared hub	Low	High	UMRA engaged from Stage 0; hub in Uganda; CRB regulations precedent; DPPA analysis in paper
Key-person dependency	Medium	High	Two engineers; documented codebase (open in this repository); research partner co-owns evaluation

13. Timeline (18 months)



14. Pathway to scale and Stage 2

Step	What	When
Stage 2 impact evaluation (≤ €1.5M)	150–300 institutions, full RCT powered on member-level outcomes, run by the research partner	Months 19–42
Sector-run hub	Hand-over plan for UMRA or UCSCU to host the aggregation server and model registry; MoU drafted in this pilot	Month 18 deliverable
Open standards	Tazama (Linux Foundation) integration so detection logic is portable	During Stage 2
Revenue model	Tiered subscription: free/low-cost for small SACCOs, cross-subsidised by MFIs and banks; pricing tested with pilot institutions at month 12	Month 12–18
Regional replication	Kenya, Tanzania, Rwanda SACCO sectors	Post-Stage 2

15. Alignment with FID selection criteria

FID criterion	How this proposal meets it
Impact on people living in poverty	Protects savings and credit access of ~18M mostly rural, low-income members; prevents institutional collapse; gender-disaggregated measurement built in.
Cost-effectiveness	Full costing against measured loss reduction; comparison with manual audit; explicit break-even analysis.
Potential for scale and sustainability	Federation hub designed as sector infrastructure (UMRA/UCSCU); open-source integration; subscription model tested in pilot; regional replicability.
Innovation	First federated-learning fraud detection for SACCOs/MFIs in Africa; solves data scarcity and privacy law together.
Readiness for Stage 1	Prototyped and validated in Stage 0 [Stage 0 result] ; ready for real-world testing at scale.
Rigorous learning design	Cluster-randomised phased roll-out, pre-registered, independent research partner; produces the estimates and power calculations needed for a Stage 2 RCT.

FID criterion	How this proposal meets it
Priority sectors	Gender-equality dimension via women's reliance on SACCOs [Stage 0 result: gender analysis] ; financial resilience of poor households.

16. Annexes to prepare

- ☐ Stage 0 final report (real-data validation, baselines, federated PoC)
- ☐ Letters of intent from ≥ 48 candidate institutions (target 60)
- ☐ Letters of support from UMRA and UCSCU
- ☐ Research partner agreement and CVs of evaluation co-PI
- ☐ Pre-analysis plan draft
- ☐ Ethics approval or submission receipt
- ☐ Data-processing agreement template (DPPA 2019 compliant)
- ☐ Alert communication and staff-fairness protocol
- ☐ Detailed budget in FID template
- ☐ Legal registration documents and financial statements of the applicant
- ☐ docs/FedFraudShield_DSA2026_Paper_v3.pdf and link to this repository